

Reality Probe: Information-Limited Physical Verification of Captured Worlds

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Captured scenes are increasingly used as executable worlds whose physical parameters can be fitted, edited, and replayed. A lower held-out observation error, however, does not by itself establish that a physical edit is more correct. We study this failure mode as deceptive physical repair: a candidate rewrite improves ordinary held-out agreement while worsening an untouched physical target. We introduce Reality Probe, an evidence-governed verification framework that separates proposal generation from independent physical evidence and permits support, veto, or unresolved outcomes. Reality Probe is the framework; Dark-Field Counterfactual Spectroscopy (DCS) is one mechanism-selective instrument tested within it. DCS constructs signed intervention contrasts that cancel lower-order common response and evaluates the residual under measurement, repeat, and numerical uncertainty. A staged falsification program shows that DCS succeeds on constructed controls but does not provide useful prospective repair-verification coverage in frozen validation: D2 resolves 0/16 truth worlds, D3 resolves 0/6, and D4V resolves 0/36 ordinary-improving repair proposals, including 14 deceptive repairs. We then freeze a new known-force compliance experiment on fresh synthetic truth worlds. This orthogonal physical channel increases median standardized evidence magnitude from 0.0488 to 0.5586, an 11.46 $\times$ gain, yet its maximum magnitude is 1.3191 against the frozen decision magnitude of 2, leaving all 36 proposals unresolved. A retrospective measured GAUGE study independently exhibits channel disagreement. These results support a narrower conclusion than a universal verifier: physical verification in the tested captured-world regimes can be information-limited, and a trustworthy system should refuse certification when the available interventions do not provide sufficient discriminating evidence.

Additional Key Words and Phrases: captured worlds, inverse physics, physical verification, counterfactual simulation, material point method, uncertainty, experiment design, Gaussian splatting

1 INTRODUCTION

Recent systems increasingly attach physical state to visual scene representations so that captured objects can be simulated, edited, and interrogated under new interactions. Physics-aware neural fields and Gaussian representations have made this coupling practical across dynamic reconstruction, material inference, and interaction synthesis [Li et al. 2023; Liu et al. 2025; Rho et al. 2026; Wada et al. 2026; Xie et al. 2024; Zhang et al. 2024]. These systems raise a verification question that is distinct from fitting: after a candidate physical rewrite improves the observations used to assess it, what evidence establishes that the rewrite is actually more correct under interventions that were not part of the repair objective?

This distinction matters because inverse physical models are commonly underdetermined by the observations available to them. Different parameterizations, transfer schemes, boundary assumptions, numerical settings, or constitutive approximations can fit similar observed motion while making different predictions elsewhere. Work on structural identifiability, optimal experimental design, model discrepancy, and simulation credibility already provides mature tools for analyzing related ambiguity [Asadi and Laksari 2025; Laudy

2026; Oberkampff et al. 2007; Wang et al. 2026b]. Our focus is narrower and operational: the object being judged is a *requested physical rewrite* of a persistent captured world, and the verifier must decide whether independent evidence supports the rewrite, contradicts it, or is insufficient to decide.

We call a repair *deceptive* when it improves an ordinary held-out observation metric while worsening an untouched physical target. This is not a claim about intent or a particular optimizer. It is a failure mode of the evidence used to approve an edit. The selected case below shows one exact example from our fresh synthetic force-compliance experiment. The repair reduces ordinary held-out error by 14.93% while increasing the untouched-target error by 9.03%. The case is selected only for visualization; aggregate statistics and decision thresholds are independent of this selection.

We investigate this problem through *Reality Probe*, an evidence-governed verification pipeline for captured executable worlds. Candidate repairs are proposed using ordinary calibration and held-out observations. Verification then applies controlled physical interventions that were not used to choose the repair, compares the truth-like reference response with candidate responses, propagates measurement, repeat, and numerical uncertainty, and produces a support, veto, or unresolved decision. Reality Probe is the contribution-level framework: it separates proposal generation from evidence, carries explicit uncertainty, preserves support/veto/unresolved outcomes, quantifies the information frontier, and permits refusal. Dark-Field Counterfactual Spectroscopy (DCS) is one mechanism-selective instrument tested inside that framework. DCS constructs signed intervention contrasts that cancel lower-order common response so that residual disagreement becomes more visible. The operators used to build these contrasts are classical finite-difference and inclusion-exclusion constructions; our contribution is their use as a repair-verification instrument and, equally importantly, the empirical characterization of where that instrument fails.

The paper follows a falsification sequence rather than a positive-only benchmark. DCS first passes a constructed positive control. It then fails to produce resolved evidence on frozen truth-world validation, adaptive witness-space refinement, and a pair-specific repair-veto population. Matched raw, Fisher-style, and maximum-motion comparators likewise provide zero resolved coverage in the repair-veto population. We therefore quantify the *information frontier*: how far the observed standardized evidence lies from the frozen decision reference. Finally, we freeze a new orthogonal channel before execution, applying known external forces and measuring compliance. This change increases median standardized signal by 11.46 $\times$ on fresh synthetic truth worlds, yet still does not cross the fixed decision magnitude.

The resulting contribution is not a claim that DCS or force compliance is a solved prospective verifier. Instead, this work contributes:



Fig. 1. Reality Probe at a glance, composed from frozen result panels. A candidate repair can improve ordinary held-out fit while worsening an untouched physical target. The repair is then asked a different physical question, its motion response is compared against the truth response, mechanism-level residuals are exposed, and the evidence is standardized. The selected case is post hoc for visualization; the 11.46× signal gain and 0/36 resolved coverage are aggregate fresh synthetic force-compliance results.

- Deceptive physical repair.** We formalize and experimentally instantiate the failure mode in which ordinary held-out fit improves while an untouched physical target worsens.
- Reality Probe.** We develop an evidence-governed rewrite-verification architecture that separates proposal generation from confirmatory physical intervention, carries explicit measurement, repeat, and numerical uncertainty, and preserves support, veto, or unresolved outcomes rather than forcing a binary verdict.
- Prospective falsification of the stronger DCS hypothesis.** We show that lower-order-annihilating intervention contrasts can succeed on a constructed control yet fail to provide useful prospective discrimination in frozen realistic regimes; this negative result is retained rather than retuned away.

- Orthogonal physical-evidence experiment.** On fresh truth worlds, changing the physical interrogation from kinematic counterfactuals to known-force compliance raises median standardized evidence by 11.46× under the unchanged decision rule, while still remaining insufficient for certification.

The central conclusion is therefore about evidence, not optimism: a captured world can look better and still be physically worse, and a verifier should remain unresolved when the available experiment does not contain enough discriminating information.

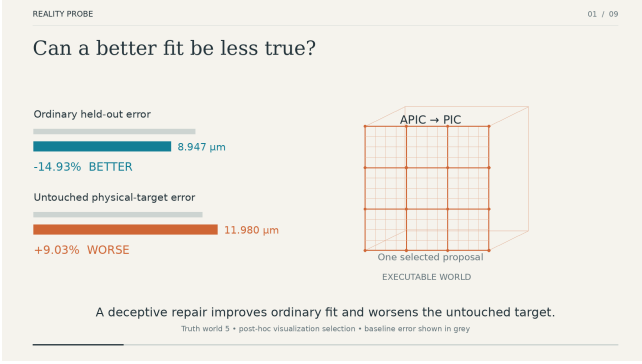


Fig. 2. Selected deceptive repair from the frozen fresh synthetic force-compliance population. The ordinary held-out error improves by 14.93% while the untouched physical-target error worsens by 9.03%. This post-hoc visualization case does not affect aggregate statistics or thresholds.

2 RELATED WORK

2.1 Physics-aware scene representations and inverse physics

PAC-NeRF couples continuum simulation to a neural radiance field for joint geometry and physical-parameter estimation from multi-view video [Li et al. 2023]. PhysGaussian integrates Material Point Method (MPM) dynamics directly with 3D Gaussian primitives [Xie et al. 2024]; subsequent work has extended physics-aware visual representations toward material initialization and refinement, interaction synthesis, and monocular or multi-material inverse physics [Liu et al. 2025; Rho et al. 2026; Wada et al. 2026; Zhang et al. 2024]. These methods establish that high-quality visual representations can support physical inference and simulation. Reality Probe does not claim novelty for Gaussian-plus-MPM simulation, parameter fitting, or differentiable inverse physics. We instead ask whether a proposed physical rewrite is independently supported once an optimizer has already produced an observational improvement.

Differentiable simulation systems such as ChainQueen and Diff-Taichi provide important infrastructure for gradient-based physical optimization [Hu et al. 2020, 2019]. In our implementation, finite differences, adjoint sensitivity, APIC/MPM replay, and material calibration are enabling components rather than contributions by themselves.

2.2 Identifiability and experimental design

Physical inference can be limited by the information content of the observed trajectories. Recent work provides structural identifiability conditions directly from video for classes of dynamical systems [Wang et al. 2026b], while measurement-grounded benchmarks such as IRIS evaluate parameter accuracy, identifiability, extrapolation, and governing-equation recovery on controlled real videos [Khanbayov et al. 2026]. In mechanics, optimal experiment design uses sensitivity, Jacobian conditioning, and Fisher-information concepts to choose informative loading conditions [Asadi and Laksari 2025]. These ideas motivate our matched baselines and the distinction between weak and informative interventions. Classical model-discrimination and optimal-experiment-design reasoning likewise

asks which excitation best separates competing explanations; Reality Probe does not claim that principle as new. Our narrower contribution is to make evidence sufficiency an explicit transaction rule for a requested captured-world rewrite, to retain unresolved outcomes, and to measure what changes when the physical evidence channel itself is changed.

2.3 Simulation credibility and measured physical fidelity

Verification, validation, uncertainty quantification, and intended-use credibility have long been central to computational modeling [Oberkampff et al. 2007]. More recent digital-twin work explicitly considers validation of simulated counterfactuals [Laudy 2026]. These frameworks make broad credibility questions explicit, but they do not remove the application-specific need to determine whether a particular physical edit is supported by the available experiment.

GAUGE provides a recent measurement-grounded benchmark for physical fidelity across simulation engines and video world models, covering rigid, cable, textile, and volumetric-deformation tasks [Wang et al. 2026a]. We use a public GAUGE foam-shearing subset only for a retrospective channel-contradiction analysis. That analysis is intentionally not presented as fresh prospective confirmation.

2.4 Measured deformable tracking

For engineering validation of the captured-world pipeline, we also use sequence C2 from the DOT deformable tracking dataset, which provides measured 3D correspondences obtained using invisible markers [Li et al. 2025]. DOT supplies geometry and motion measurements, but not the stress-free state, loads, thickness, mass, or constitutive ground truth required to identify the true cloth material. We therefore treat the inferred parameters in that benchmark as model-conditioned effective quantities and keep this measured-source system check separate from the main verification claim.

3 PROBLEM FORMULATION

Let M denote an executable captured world and let R_0 and R_1 denote a baseline and a proposed repaired physical realization. We distinguish three data roles: fitting evidence used to calibrate or propose a rewrite, ordinary held-out evidence used to measure generalization within the same observation family, and an untouched target or independent physical intervention used only after the repair is proposed.

Let $L_{obs}(R)$ be the ordinary held-out loss and $L_{target}(R)$ be an independent physical target. We define a deceptive repair by

$$L_{obs}(R_1) < L_{obs}(R_0) \quad \text{and} \quad L_{target}(R_1) > L_{target}(R_0).$$

This definition is deliberately agnostic to why the mismatch occurs. Model-form inadequacy, parameter compensation, transfer choice, boundary assumptions, and numerical artifacts can all create observationally plausible but physically different worlds.

A verification channel produces an error $e(R, Y)$ against independent evidence Y and an uncertainty scale σ . We use the signed progress statistic

$$z = \frac{e(R_0, Y) - e(R_1, Y)}{\sigma}.$$

The frozen decision rule is

$$z \geq 2 \Rightarrow \text{support}, \quad z \leq -2 \Rightarrow \text{veto}, \quad |z| < 2 \Rightarrow \text{unresolved}.$$

The magnitude 2 is an operational decision reference fixed before the fresh evaluation; it is not claimed to be a universal physical constant. The distinction matters because the main experiments remain unresolved rather than being reclassified after inspection.

4 REALITY PROBE

4.1 Captured-world rewrite and verification pipeline

Reality Probe separates proposal generation from verification. A captured bundle provides appearance-linked state, particle state, observations, uncertainty sidecars, and provenance. The implementation calibrates a candidate physical model using the fitting split, evaluates ordinary held-out error, proposes a local or model-level rewrite, and only then evaluates independent counterfactual evidence. If the verification channel produces support, the system may commit the rewrite; if it produces veto, the rewrite is rejected; if evidence is unresolved, the system refuses certification rather than forcing a binary answer.

The evidence layer deliberately has three outcomes, while the lower-level world transaction has two terminal states: committed or not committed. In the implementation, support can lead to a verified commit; both veto and unresolved map to a rejected/non-commit transaction, and any tentative mutation is rolled back. The three-way scientific distinction is retained in the evidence record even though the storage transaction is binary.

The verification transaction is deliberately sequential: proposal generation is completed before the independent physical probe is evaluated, and the resulting evidence may support, veto, or leave the rewrite unresolved. This ordering prevents the confirmatory channel from silently becoming part of the proposal objective.

The research implementation is in C++20 and uses nonlinear MPM replay, APIC/PIC transfer variants, finite-difference and adjoint influence paths, deterministic experiment drivers, frozen JSON/CSV result ledgers, and SHA-256-indexed evidence packaging. The paper does not depend on native rendering for its quantitative conclusions; Metal and Vulkan render paths are used for system integration and presentation.

4.2 Dark-Field Counterfactual Spectroscopy

For a model M and physical intervention u , let $F_M(u)$ denote an observable response vector. DCS applies a signed intervention measure μ to form

$$\mathcal{D}_\mu[F_M] = \sum_i w_i F_M(u_i).$$

For a declared mechanism order k , the stencil weights are required to annihilate all lower-order multivariate monomials,

$$r_\alpha = \sum_i w_i u_i^\alpha \approx 0, \quad |\alpha| < k.$$

The implementation enumerates these monomials and validates the maximum residual against a tolerance rather than hard-coding only one stencil. A familiar order-2 one-dimensional example is

$$F(+a) + F(-a) - 2F(0). \quad (1)$$

which cancels constant and linear response. In the experiments, however, the order-2 comparator is synthesized on a 3×3 intervention lattice; Eq. 1 is explanatory rather than the literal selected stencil.

The system also implements finite-amplitude Boolean-lattice interactions through the Möbius contrast

$$\kappa(S) = \sum_{T \subseteq S} (-1)^{|S|-|T|} F(T).$$

We treat this as an interaction statistic, not as a newly derived mixed-derivative theorem.

4.3 Uncertainty and standardized separation

The evidence denominator separates measurement, repeat, and numerical uncertainty. Independent measurement and repeat variance propagate through a signed stencil with the squared ℓ_2 weight gain $\sum_i w_i^2$. Numerical error is treated more conservatively because discretization error can be correlated across intervention points. D3 additionally estimates numerical uncertainty directly in witness space by evaluating the same signed witness at nominal and refined solver resolution.

For candidate pairs, the verifier evaluates worst-case standardized separation over the retained response bundle. Active stencil synthesis first projects candidate disagreement into the nullspace of the lower-moment constraints and then searches deterministic projected directions. This is a heuristic for a non-convex maximin problem; the implementation does not provide a global-optimality certificate.

4.4 Known-force compliance as orthogonal information

After D4V, we freeze a separate experiment to test whether the low evidence magnitude is specific to the DCS construction or reflects a broader shortage of informative physical observations. The new channel leaves the fitting and repair-proposal path unchanged and applies a known force only after a candidate repair has been selected.

Six fresh synthetic truth worlds use APIC transfer with Young's modulus in $\{14125, 16625, 18125\}$ Pa and Poisson ratio in $\{0.22, 0.32\}$, with truth timestep 2.5×10^{-5} s. Candidate families are fixed before execution. The bottom particle layer is kinematically fixed, gravity is zero, and each top-layer particle receives 40 N in each of four experiments: $+x$, $-x$, $+y$, and $+z$. For every direction we record the mean top-layer and interior-layer displacement in three axes, yielding a six-component response per force experiment.

Numerical uncertainty is the RMS difference between nominal and half-timestep candidate response bundles. Measurement and repeat scales are each frozen at 2×10^{-5} m per component. The force progress statistic is

$$z_{\text{force}} = \frac{e(B, Y) - e(R, Y)}{\sqrt{\sigma_{\text{meas}}^2 + \sigma_{\text{repeat}}^2 + \sigma_{\text{num}, B}^2 + \sigma_{\text{num}, R}^2}}.$$

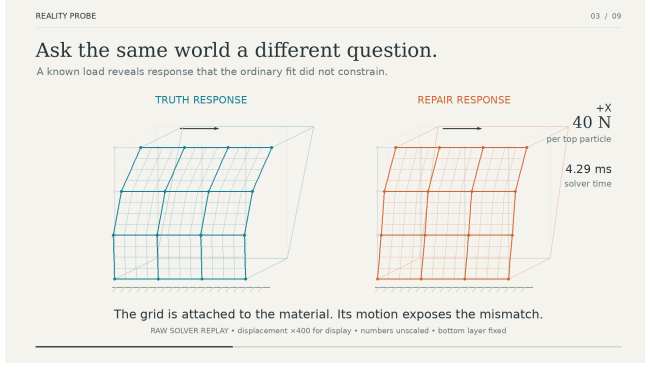


Fig. 3. Same controlled +X force probe applied to truth and repaired worlds. The displayed deformation is magnified 400x only for legibility; reported numbers remain unscaled.

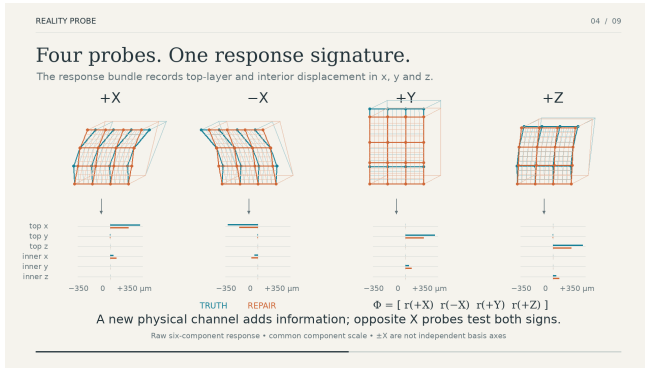


Fig. 4. Four-direction force-response fingerprint for the selected frozen case. The six-component response bundle records top-layer and interior displacement under +X, -X, +Y, and +Z probes; truth and repair are plotted on a shared component scale.

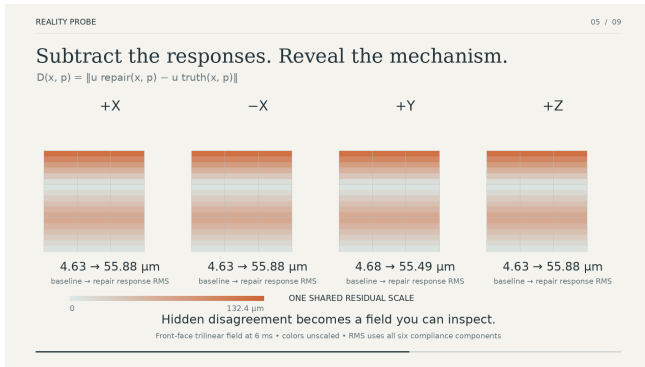


Fig. 5. Solver-driven residual field for the same frozen visualization case. All four force directions share one residual scale; the field is explanatory and is not itself the DCS decision statistic.

The same decision magnitude of 2 is used for the DCS comparator and force channel. Force amplitude, directions, truth worlds, candidate grid, and threshold are frozen before execution and are not returned after observing the results.

5 EXPERIMENTAL PROGRAM

Our experiments are organized to distinguish implementation sanity checks from frozen validation, discovery analyses, retrospective measured evidence, and fresh follow-on evidence. Table 1 summarizes the central outcomes.

Main experimental stages. “Crosses frozen boundary” means that the standardized evidence reaches the predeclared support/veto magnitude; zero crossings do not imply that an experiment produced no scientific information.

5.1 Graphics-facing benchmark and visual evidence coverage

A graphics submission needs recognizable benchmark evidence and inspectable visual consequences, not only an internal systems ledger. We therefore make the evidence surface explicit rather than asking the reader to infer it from the implementation. Table 2 separates public measured datasets, controlled synthetic evaluation, and visualization-only geometry. This classification does not change any experimental gate or elevate retrospective evidence into prospective confirmation.

The paper’s deceptive-repair, same-probe, force-fingerprint, residual-field, signal-gain, and information-limit panels are generated from committed solver trajectories or frozen result ledgers. They are intended to show what the tested physical differences look like, while the numerical conclusions continue to come from the JSON/CSV evidence records. In other words, the visual layer exposes the consequence of the experiments without becoming a second, informal source of truth.

5.2 D1: constructed positive control

D1 verifies the execution path under deliberately constructed deceptive cases. All 64 proposed repairs improve the ordinary criterion and are deceptive by construction. DCS rejects all 64. Median ordinary observation improvement is 60.88%, and the median witness-degradation ratio is 9.375. We use this stage only as an implementation control; it does not establish prospective generalization.

5.3 D2: active selection and frozen validation

An active-selection discovery comparison evaluates how well different witnesses rank target behavior. On D2b, ranking agreement is 95.83% for raw maximin, 87.50% for Fisher/Jacobian selection, 75.00% for the same-cost raw bundle, 66.67% for maximum motion, 62.50% for DCS order-2 maximin, and 45.83% for random selection. DCS therefore does not dominate simpler matched baselines.

The frozen D2 validation uses 16 truth worlds. No world reaches the decision magnitude. Median standardized separation is 0.041578, median numerical RMS is 6.469×10^{-6} m, and the maximum moment residual is 3.886×10^{-16} . The stencil algebra is therefore numerically well satisfied while the physical evidence remains weak.

5.4 D3: witness-space refinement and adaptive order

D3 tests whether direct witness-space numerical treatment or higher mechanism order recovers useful separation. Six fresh synthetic truth worlds are evaluated. Adaptive selection chooses order 2 in all six cases and never chooses order 3. Median adaptive separation

Table 1. Main experimental stages. “Resolved” refers to the frozen support/veto decision rule, not whether the scientific question was informative.

Stage	Population	Crosses frozen boundary	Main outcome
D1	64 repairs	64/64	Constructed positive control; median ordinary improvement 60.88%, median witness degradation ratio 9.375.
D2	16 worlds	0/16	Median standardized separation 0.041578.
D3	6 worlds	0/6	Median adaptive separation 0.0588583; lower numerical floor does not recover observability.
D4V	36 repairs	0/36	14 deceptive and 22 beneficial; matched verification channels also have zero resolved coverage.
GAUGE	10 trials	–	Retrospective measured channel contradiction; longitudinal endpoint preferred in 9/10 held-out repeats.
OFC	36 repairs	0/36	12 deceptive and 24 beneficial; known-force compliance increases evidence scale but not coverage.

Table 2. Graphics-facing evidence coverage. Public measured benchmarks are kept distinct from controlled synthetic partitions and from visualization-only geometry.

Source	Evidence class	What it contributes	Claim boundary
DOT C2 [Li et al. 2025]	Public measured trajectories	End-to-end captured-world ingestion, fit/held-out replay, local rewrite, and rollback on measured motion.	Stress-free state, loads, thickness, mass, and constitutive ground truth are not supplied; fitted material values remain model-conditioned.
GAUGE foam shearing [Wang et al. 2026a]	Public measured benchmark	Ten held-out repeats for a measured channel-contradiction analysis.	Retrospective/post-hoc evidence only; it is not fresh prospective confirmation of DCS.
D2/D3/D4V + OFC	Frozen controlled synthetic	Known-ground-truth falsification, pair-specific repair evaluation, and orthogonal force-compliance observability under a frozen decision rule.	Controlled evidence, not a measured-world benchmark.
Stanford Bunny visual suite	Public geometry visualization carrier	Solver-driven deformation, force direction, mechanism fingerprints, and residual fields for qualitative inspection.	Visualization only; the bunny is not benchmark geometry and contributes no quantitative result.

is 0.0588583. Refinement lowers the median numerical RMS from 8.435×10^{-7} m for order 2 to 2.153×10^{-7} m for order 3, yet no truth world is resolved. Ranking agreement remains modest: Fisher 61.11%, raw bundle 58.33%, maximum motion 58.33%, pair-aware raw 55.56%, adaptive DCS 52.78%, and order-3 DCS 47.22%.

This stage separates two issues that are often conflated: numerical witness fidelity can improve without making the experiment more informative about the competing physical worlds.

5.5 D4V: pair-specific repair veto

D4V evaluates 36 repairs that already improve ordinary held-out error. Fourteen are deceptive according to the untouched target and 22 are beneficial. DCS, raw-bundle, pair-specific raw, Fisher, and maximum-motion verification all produce zero resolved coverage at the inherited $|z| \geq 2$ decision rule. The numeric table below reports the D4V and fresh follow-on proposal counts, followed by the fresh proposal landscape.

A post-hoc information-frontier analysis quantifies the gap. In D4V the median DCS magnitude is 0.05024 against the frozen magnitude 2 decision reference, corresponding to roughly 39.8x median

Table 3. Ordinary-improving proposal populations. D4V and fresh synthetic OFC are independently generated partitions and their counts must not be mixed.

Population	Deceptive	Beneficial	Resolved
D4V	14	22	0/36
Fresh OFC	12	24	0/36

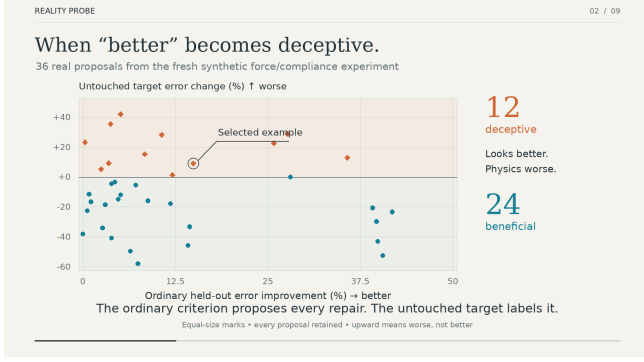


Fig. 6. Fresh orthogonal force-compliance proposal landscape. The figure shows where ordinary held-out improvement and the untouched physical target disagree across the fresh population.

signal amplification if uncertainty were otherwise unchanged. The matched channels are also far from resolved coverage. We treat this analysis as diagnostic only; it does not modify any frozen gate.

5.6 Measured-source engineering check

The end-to-end system is also exercised on measured DOT C2 trajectories [Li et al. 2025]. The importer keeps measured 3D correspondences separate from explicit model proxies for rest state, density, thickness, mass, Gaussian photometry, and uncertainty. Fit-only calibration and held-out replay produce a model-conditioned effective material setting rather than a claim of true cloth-property recovery. A proposed local material rewrite is rejected and rolled back because the complete independent derivative-verification contract is not met, despite satisfying the configured nonlinear linearization-error tolerance. This result validates the commit/refusal mechanism on measured-source data but is not used as evidence for the central deceptive-repair claim.

The same pinned DOT C2 source is also exercised by the native Vulkan measured-world showcase, which emits baseline and roll-back hero views, close-ups, a comparison plate, and an eight-view turntable. Because the candidate rewrite is rejected, the final displayed state intentionally returns to the baseline state. These images are qualitative implementation evidence for the measured captured-world path; they are not additional evidence that the fitted material parameters are physically unique or ground-truth accurate.

5.7 GAUGE: retrospective measured channel contradiction

We analyze ten held-out foam-shearing repeats from GAUGE [Wang et al. 2026a]. The committed paired-diagnostics CSV is channel-aggregated: each row reports win/loss/tie counts over these same

Table 4. Public DOT C2 measured-source engineering check. The material parameters are model-conditioned effective values, not material ground truth.

Quantity	Result
Measured correspondence points	225
Trajectory observations (fit / held-out)	675 (585 / 90)
Effective E, ν	7.5 kPa, 0.45
Dynamic RMS (fit / held-out)	4.391 / 4.417 mm
Adaptive candidate set	182/225 particles; 94.80% abs. gradient
Rewrite verification	nonlinear pass; oracle fail; rollback



Fig. 7. Held-out DOT C2 measured-versus-predicted geometry at the later validation checkpoint ($t = 0.167$ s), shown in the camera-frame x/y projection. Circles are measured correspondences, crosses are Vulkax predictions, and connecting segments are true-scale residuals rather than magnified arrows. Across the 90 held-out samples from both nonzero checkpoints, the 3-D RMS is 4.417 mm. This evaluates the model-conditioned captured-world replay; DOT C2 does not provide material ground truth.

ten repeats rather than representing a single trial. Ordinary face-plus-marker metrics prefer the finite-overlap candidate in all 10 repeats. Marker dark-field also prefers overlap in all 10. In contrast,

Table 5. GAUGE retrospective channel comparison. Lower error is better. The sign test is post hoc.

Metric	Endpoint	Overlap
Median marker error (m)	0.002428	0.001709
Median longitudinal error	0.003131	0.005584
Longitudinal held-out wins	9/10	1/10

Table 6. Fresh synthetic orthogonal force-compliance results.

Quantity	DCS	Force
Median $ z $	0.04877	0.55863
Maximum $ z $	0.17279	1.31911
Resolved coverage	0/36	0/36
Median factor to $ z = 2$	41.01×	3.58×
Best-case factor to $ z = 2$	11.57×	1.52×

the longitudinal mechanism channel prefers the endpoint model in 9 of 10 repeats. Median marker error is 0.002428 m for endpoint and 0.001709 m for overlap, whereas median longitudinal error is 0.003131 for endpoint and 0.005584 for overlap. A post-hoc exact two-sided sign test on the longitudinal direction yields $p = 0.02148$.

The metric conflict was known before DCS was formulated. We therefore label this evidence retrospective throughout and do not use it as prospective confirmation.

5.8 Fresh orthogonal force-compliance study

The fresh synthetic force experiment yields 36 ordinary-heldout-improving proposals: 12 deceptive and 24 beneficial. Fresh DCS median $|z|$ is 0.04877 with maximum 0.17279. Force compliance median $|z|$ is 0.55863 with maximum 1.31911. The ratio of aggregate medians is 11.455. Despite the substantial increase, DCS resolves 0/36 and force compliance resolves 0/36. The median additional amplification needed for the force channel to reach magnitude 2 is 3.58x; the best case still requires

1.52x.

Decision-threshold robustness. The primary decision magnitude remains the prospectively frozen $\tau = 2$. To check whether the central negative conclusion is merely an artifact of that particular number, we also inspect nearby thresholds without changing the primary rule. The frozen per-proposal OFC visualization ledger retains the complete signed DCS and force progress scores, so the diagnostic sweep can be reported exactly. Fresh DCS crosses neither support nor veto even at $\tau = 1$. At $\tau = 1$, force compliance resolves 14/36 proposals: 6 support and 8 veto. Five of the 12 deceptive repairs are vetoed (41.7% recall), while 3 of 24 beneficial repairs are falsely vetoed (12.5%). The result therefore remains below the preregistered advancement target even under this looser post-hoc threshold. At $\tau = 1.5$ and above, force coverage returns to 0/36. This robustness check leaves the primary conclusion unchanged: under the frozen $\tau = 2$ rule the fresh population remains uncertified, and the observed transition between $\tau = 1$ and $\tau = 1.5$ makes the threshold dependence explicit rather than hiding it.

Table 7. Threshold-robustness audit from the frozen per-proposal ledger. The $\tau = 2$ row is the prospectively frozen primary rule; other rows are post-hoc sensitivity checks and do not redefine the decision policy.

Decision magnitude τ	DCS crossings	Force crossings
1.0	0/36	14/36
1.5	0/36	0/36
2.0	0/36	0/36
2.5	0/36	0/36
3.0	0/36	0/36

The ratio of the aggregate median evidence magnitudes is 11.455x. The full post-hoc sweep is generated from the frozen per-proposal ledger by `research/analysis/threshold_sensitivity_ofc.py` and stored as `research/paper_data/THRESHOLD_SENSITIVITY_OFC_2026-09-21.csv`.

The protocol includes an advancement gate requiring at least 30% resolved coverage, at least 50% veto recall over deceptive repairs, no more than 25% false veto rate over beneficial repairs, better coverage than DCS, higher median magnitude than DCS, and finite numerical refinement for every proposal. The force experiment satisfies the population-size, false-veto, median-signal, and numerical checks but fails the coverage and recall requirements. The result is therefore retained as a negative gate with a strong standardized-evidence increase, not relabeled as a successful verifier.

6 RESULTS AND ANALYSIS

6.1 Deceptive repair is a distinct failure mode

The D4V and fresh synthetic OFC populations establish that lower ordinary held-out error does not uniquely determine whether the untouched physical target improves. In D4V, 14 of 36 ordinary-improving repairs are deceptive. In the fresh synthetic OFC population, 12 of 36 are deceptive. These counts are not intended as estimates of a universal deception rate; they establish that the failure mode occurs under controlled candidate-generation pipelines and must therefore be considered by any verification rule that approves physical edits from observation fit alone.

6.2 The stronger DCS hypothesis is not supported in the tested regime

The constructed D1 control demonstrates that the implementation can produce a strong dark-field rejection when the experiment is engineered to contain that distinction. The subsequent stages do not preserve that strength. D2 and D3 remain far below the decision magnitude, DCS does not dominate matched ranking baselines, and D4V provides zero resolved coverage. These negative results are central rather than incidental: they rule out the stronger interpretation that mechanism-selective cancellation alone is sufficient for repair certification in the tested regime.

The D3 numerical refinement result is particularly informative. Reducing the witness-space numerical floor by several-fold does not materially increase standardized separation. The limiting factor is therefore not simply solver discretization error in the original witness evaluation.

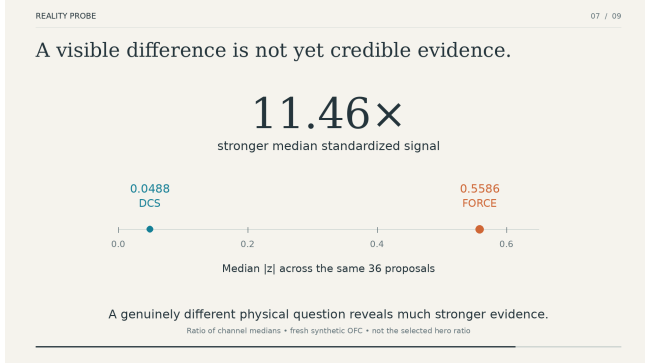


Fig. 8. Fresh known-force compliance raises the median standardized signal by 11.46x relative to fresh DCS without changing the frozen decision rule.

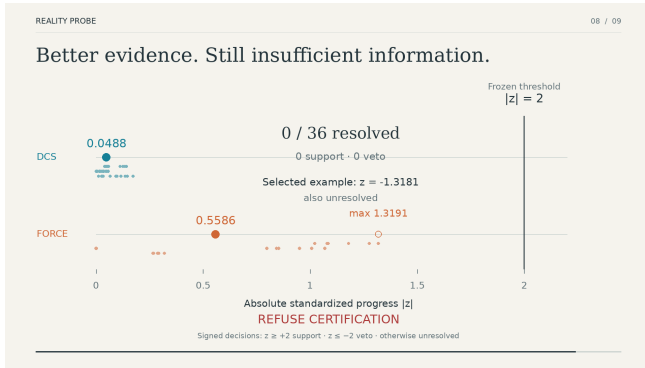


Fig. 9. Information-limit summary from the frozen fresh synthetic orthogonal-information experiment. Known-force compliance increases standardized evidence substantially, but the strongest case remains below the frozen decision magnitude.

6.3 Orthogonal physical information changes the evidence scale

The information-limit result below summarizes the frozen evidence scale. The known-force compliance channel moves the median from the same order as the frozen D2/D3/DCS evidence to 0.5586. This is an order-of-magnitude change in the observed standardized evidence scale under a genuinely different physical interrogation, not a threshold adjustment or a new score applied to the same observations. We interpret this as strong evidence that observability depends on the experiment, while avoiding the stronger claim that information content is the unique limiting cause.

This result supports the interpretation that verification quality depends strongly on which physical question is asked. Kinematic observations that are adequate for proposing a plausible repair need not be adequate for discriminating that repair from alternatives. A force-controlled experiment adds information that is not available to the fitting objective.

6.4 Stronger evidence is not sufficient evidence

The strongest result in the fresh synthetic force population is $|z| = 1.3191$, still below the decision magnitude. The selected visual

example has signed force progress $z = -1.3181$: evidence moves substantially in the veto direction but remains unresolved. The system therefore refuses certification. We view this behavior as preferable to lowering the threshold after observing the partition or interpreting an intermediate score as a definitive physical verdict.

The information-frontier view also avoids a misleading binary narrative. The force channel is not equivalent to DCS simply because both return 0% resolved coverage. Their evidence magnitudes are materially different, and the remaining gap to the decision rule is correspondingly different. The experiment therefore supplies a quantitative direction for future sensor and intervention design even though it does not support deployment-level verification.

7 DISCUSSION

7.1 What is established

The experiments support three linked conclusions. First, observationally improving physical repairs can be deceptive with respect to an untouched target. Second, mechanism-sensitive verification signals can remain too weak to support a reliable repair decision even when their numerical implementation is stable. Third, changing to a genuinely different physical channel can substantially increase observability without necessarily making the evidence sufficient for certification.

These conclusions are narrower than a universal physical-correctness claim, but they are operational. A persistent captured-world system can preserve the distinction between fitting, verification, and unresolved evidence, and it can refuse a requested rewrite without treating refusal as a pipeline failure.

7.2 Relation to identifiability and experiment design

The force-compliance result is consistent with the broader principle that parameter and model discrimination depend on excitation and measurement design [Asadi and Laksari 2025; Wang et al. 2026b]. Our experiments do not introduce that principle. Their contribution is to expose its consequence inside a physical-rewrite workflow: verification can fail not because a candidate looks implausible, but because the available experiment does not separate plausible candidates strongly enough. This distinction changes the system response from “pick the best score” to “request more informative evidence or remain unresolved.”

7.3 Relation to simulation verification and digital twins

Simulation credibility frameworks already separate code verification, solution verification, model validation, and uncertainty [Oberkamp et al. 2007]. Digital-twin validation work similarly emphasizes which counterfactual claims are supported by observable validation data [Laudy 2026]. Reality Probe should therefore be read as an application-specific mechanism rather than a replacement for those frameworks. Its scope is a physically editable captured world where a candidate rewrite is proposed and an independent intervention is used to decide whether the rewrite should be committed.

7.4 Limitations

The main limitations are substantive.

- **Fresh prospective force evidence is synthetic.** The orthogonal force-compliance experiment uses controlled MPM truth worlds rather than a measured force-sensor setup. A real prospective force experiment is required for a stronger measured-world claim.
- **GAUGE is retrospective.** The measured channel contradiction motivates the problem and demonstrates that different observables can disagree, but it is not an untouched prospective test of DCS or force compliance.
- **The physical regime is limited.** The principal experiments use a fixed deformable MPM setup and selected APIC/PIC candidate families. Generalization across constitutive families, contact regimes, and solvers is not established.
- **The decision magnitude is operational.** The value 2 is frozen before evaluation and prevents post-hoc threshold tuning, but it is not a universal law for all physical verification tasks. A post-hoc robustness audit uses the frozen per-proposal ledger: force compliance has 14/36 crossings at $\tau = 1$ (6 support and 8 veto), while both fresh channels have zero crossings for every $\tau \geq 1.5$.
- **DCS synthesis is heuristic.** Lower-moment annihilation is exact up to tolerance, but the projected maximin search is not globally optimal.
- **The response-jet interpretation is operational.** The implementation estimates order of contact from validated witnesses; it does not reconstruct a complete symbolic Taylor tensor.
- **The selected hero case is post hoc.** It is chosen deterministically for explanation and does not influence thresholds, aggregate statistics, or the standardized-evidence result.
- **Spatial residuals are explanatory.** Solver-driven residual fields localize response disagreement but are not themselves the DCS decision statistic.
- **Measured DOT parameters are model-conditioned.** The DOT benchmark lacks stress-free state, loads, thickness, mass, and constitutive ground truth for C2; fitted material values are therefore not presented as true cloth properties.

7.5 Future work

The most direct next step is not another retuning of the frozen DCS or force partitions. It is a separately frozen measured experiment with an independent force sensor or another calibrated excitation that materially changes the available physical information. Modal or frequency-domain excitation, richer constitutive families, and intervention-specific sensor design are plausible directions. Any such experiment should keep proposal generation blind to the confirmatory channel and should preserve a predeclared support/veto/unresolved rule.

8 REPRODUCIBILITY AND IMPLEMENTATION NOTES

The repository records the scientific freeze dated 2026-09-21 at commit a9da8c0aa868. The one-command runner builds and tests the code, executes the staged experiments, regenerates deterministic figures and tables, validates frozen numerical values, captures compiler/system/GPU provenance, and assembles a SHA-256-indexed

evidence archive. Scientific conclusions are read from machine-readable JSON ledgers rather than from rounded figure labels.

The core DCS implementation is separated from experiment drivers and includes arbitrary-order lower-moment validation, signed response application, finite-amplitude counterfactual cumulants, standardized discrepancies, uncertainty propagation, active stencil synthesis, witness-space numerical refinement, and spatial localization. The orthogonal force protocol and fresh synthetic truth worlds are versioned separately from the earlier D2/D3/D4V partitions. This separation is used to prevent a negative gate from being silently replaced by a retuned result on the same data.

Generative-AI-assisted research workflow. OpenAI ChatGPT was used during research for code drafting and debugging, experiment orchestration and analysis scripts, visualization/manuscript tooling, and manuscript revision. AI-generated material was not treated as empirical evidence. Experimental partitions, thresholds, executed outputs, numerical claims were checked against committed research artifacts, and cited sources were checked against publication records before inclusion in the manuscript.

Paper-facing visualizations are deterministic. Solver-driven surfaces may use an explicit display magnification to make micrometer-scale motion visible; reported numerical values are unscaled. The Stanford Bunny, where used in presentation assets, is a visualization carrier rather than benchmark geometry and requires source attribution. The companion mathematical explainer uses the actual solver body and does not require the bunny.

9 CONCLUSION

A physically editable captured world should not equate better observation fit with better physics. Across controlled repair populations, we find ordinary-improving repairs that worsen an untouched physical target. Reality Probe provides the evidence-governed transaction around such repairs; DCS is one tested mechanism-selective instrument inside it. Frozen D2, D3, and D4V experiments do not support a claim that DCS is a general prospective verifier. A separately frozen known-force compliance channel increases median standardized evidence by 11.46x on fresh synthetic truth worlds, yet still leaves every proposal below the decision magnitude. The resulting lesson is practical: in the tested regimes, physical verification can be constrained by the information supplied by the experiment, and a trustworthy system should explicitly refuse certification when that evidence is insufficient.

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